1202SCG Calculus 1
1020ENG Engineering Mathematics 2

Week 2 Lecture

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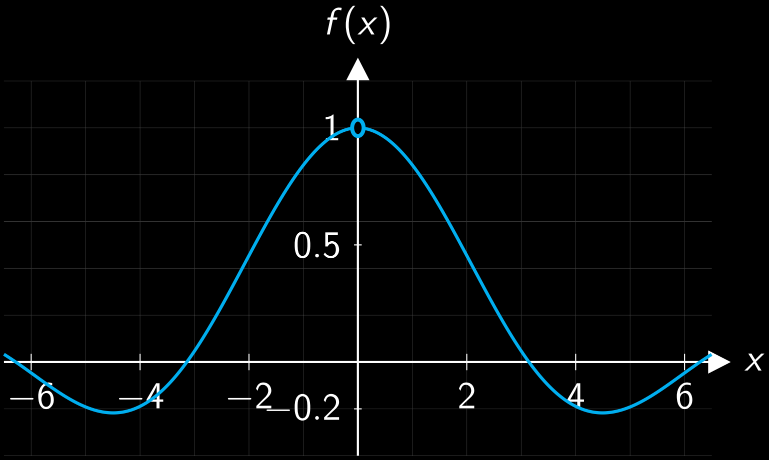
What Happens as $x \to 0$?

$$f(x) = \frac{\sin(x)}{x}$$



Formal Definition of a Limit

We say

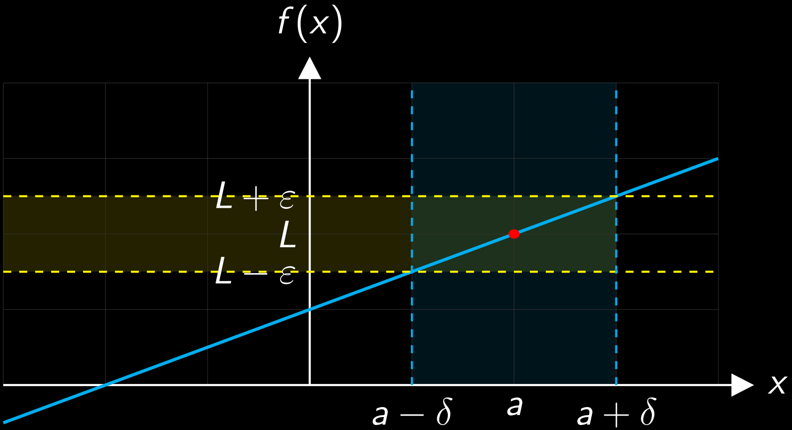
$$\lim_{x \to a} f(x) = L$$

if, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

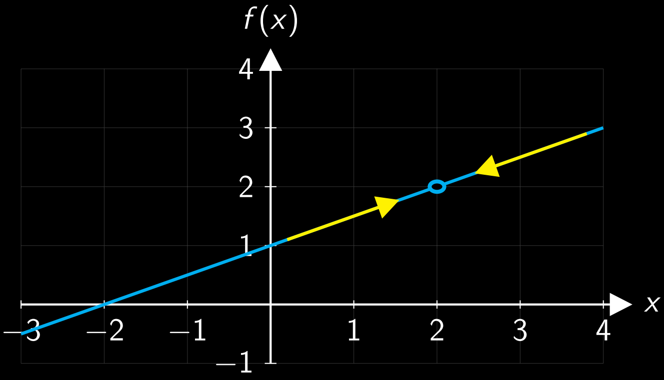
The ε–δ Picture

$$\lim_{x \to a} f(x) = L$$



An Informal View of a Limit

Imagine walking along the graph toward $x = 2$.
If you took one more step, what value would you expect the function to take?



$$\lim_{x \to 2} f(x) = 2.$$

Calculating a Limit Numerically

For $f(x) = \dfrac{\sin(x)}{x}$, calculate values close to $x = 0$.

$x < 0$	-0.5	-0.1	-0.05	-0.01
$f(x)$	0.95885	0.99833	0.99958	0.99998

$x > 0$	0.5	0.1	0.05	0.01
$f(x)$	0.95885	0.99833	0.99958	0.99998

The values from both sides suggest $\displaystyle\lim_{x \to 0} \frac{\sin(x)}{x} = 1$.

Numerical Limits Need Both Sides

For

$$f(x) = \frac{x^2 - 4}{x - 2},$$

calculate values close to $x = 2$.

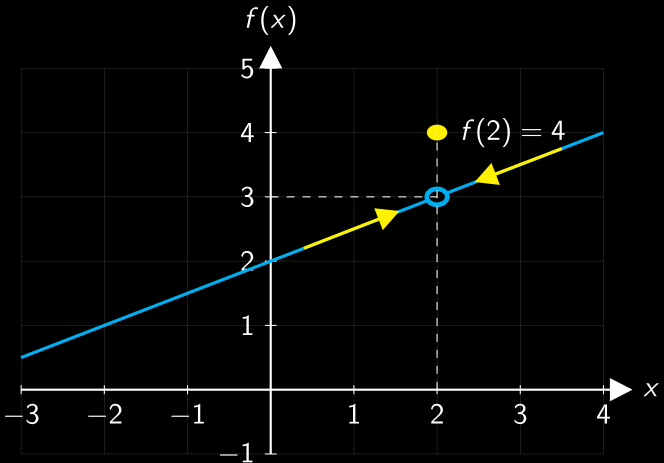
$x < 2$	1.5	1.9	1.99	1.999
$f(x)$	3.5	3.9	3.99	3.999

$x > 2$	2.5	2.1	2.01	2.001
$f(x)$	4.5	4.1	4.01	4.001

The values from both sides suggest

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4.$$

Approaching from Both Sides



$$\lim_{x\to 2} f(x) = 3, \text{even though} f(2) = 4.$$

A Limit by Factorisation

Evaluate

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}.$$

Direct substitution gives $0/0$, an **indeterminate form**.

$$\begin{aligned}
\lim_{x \to 2} \frac{x^2 - 4}{x - 2} &= \lim_{x \to 2} \frac{(x-2)(x+2)}{x - 2} \\
&= \lim_{x \to 2} (x + 2) \\
&= 4.
\end{aligned}$$

Another Factorisation Example

Evaluate

$$\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2}.$$

$$\begin{aligned} \lim_{x \to 2} \frac{x^2 + x - 6}{x - 2} &= \lim_{x \to 2} \frac{(x+3)(x-2)}{x-2} \\ &= \lim_{x \to 2} (x+3) \\ &= 5. \end{aligned}$$

A Third Factorisation Example

Evaluate

$$\lim_{x\to 3} \frac{x^2-9}{x^2-5x+6}.$$

$$\begin{aligned}
\lim_{x\to 3} \frac{x^2-9}{x^2-5x+6} &= \lim_{x\to 3} \frac{(x-3)(x+3)}{(x-3)(x-2)} \\
&= \lim_{x\to 3} \frac{x+3}{x-2} \\
&= \frac{3+3}{3-2} \\
&= 6.
\end{aligned}$$

As $x \to \infty$

The denominator grows without bound, so each reciprocal approaches zero:

$$\lim_{x \to \infty} \frac{1}{x} = 0,$$

$$\lim_{x \to \infty} \frac{1}{x^2} = 0,$$

$$\lim_{x \to \infty} \frac{1}{x^3} = 0,$$

$$\lim_{x \to \infty} \frac{1}{x^4} = 0.$$

The same applies as $x \to -\infty$.

Limits as $x \to \infty$

Evaluate

$$\lim_{x\to\infty} \frac{3x^2 - x + 2}{x^2 + 4}.$$

Divide every term in the numerator and denominator by x^2:

$$\begin{aligned}
\lim_{x\to\infty} \frac{3x^2 - x + 2}{x^2 + 4} &= \lim_{x\to\infty} \frac{\frac{3x^2}{x^2} - \frac{x}{x^2} + \frac{2}{x^2}}{\frac{x^2}{x^2} + \frac{4}{x^2}} \\
&= \lim_{x\to\infty} \frac{3 - \frac{1}{x} + \frac{2}{x^2}}{1 + \frac{4}{x^2}} \\
&= \frac{3 - 0 + 0}{1 + 0} \\
&= 3.
\end{aligned}$$

Limits as $x \to -\infty$

Evaluate

$$\lim_{x \to -\infty} \frac{-2x^3 + x}{x^3 + 4}.$$

Divide every term in the numerator and denominator by x^3:

$$\begin{aligned}
\lim_{x \to -\infty} \frac{-2x^3 + x}{x^3 + 4} &= \lim_{x \to -\infty} \frac{\frac{-2x^3}{x^3} + \frac{x}{x^3}}{\frac{x^3}{x^3} + \frac{4}{x^3}} \\
&= \lim_{x \to -\infty} \frac{-2 + \frac{1}{x^2}}{1 + \frac{4}{x^3}} \\
&= \frac{-2 + 0}{1 + 0} \\
&= -2.
\end{aligned}$$

Limits as $x \to \infty$

Evaluate

$$\lim_{x\to\infty} \frac{x^2+1}{x+1}.$$

Divide every term in the numerator and denominator by x:

$$\lim_{x\to\infty} \frac{x^2+1}{x+1} = \lim_{x\to\infty} \frac{\frac{x^2}{x}+\frac{1}{x}}{\frac{x}{x}+\frac{1}{x}}$$

$$= \lim_{x\to\infty} \frac{x+\frac{1}{x}}{1+\frac{1}{x}}$$

The expression grows without bound, so the limit does not exist.

One-Sided Limits

A left-hand limit only uses values with $x < a$:

$$\lim_{x \to a^-} f(x) = L_{\text{left}}.$$

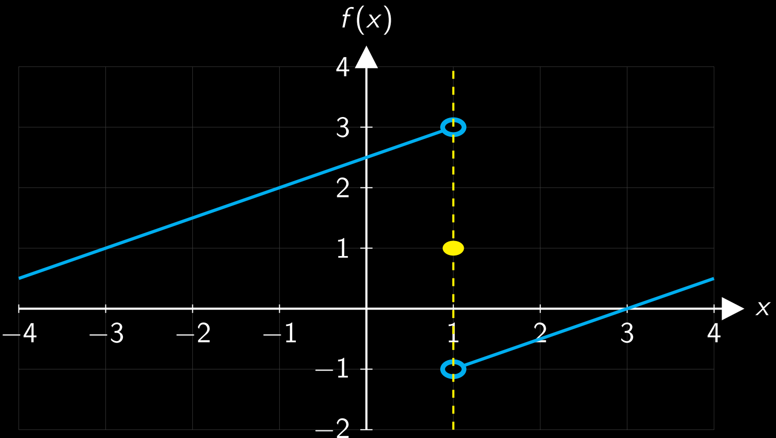
A right-hand limit only uses values with $x > a$:

$$\lim_{x \to a^+} f(x) = L_{\text{right}}.$$

The two-sided limit exists only when

$$L_{\text{left}} = L_{\text{right}}.$$

When One-Sided Limits Disagree



$$\lim_{x \to 1^-} f(x) = 3, \qquad \lim_{x \to 1^+} f(x) = -1.$$

$$\therefore \quad \lim_{x \to 1} f(x) \text{ does not exist.}$$

Formal Definition of Continuity

A function f is **continuous at** $x = a$ if

$$\lim_{x \to a} f(x) = f(a).$$

Equivalently, all three conditions must hold:

1. $f(a)$ is defined.
2. $\lim_{x \to a} f(x)$ exists.
3. $\lim_{x \to a} f(x) = f(a)$.

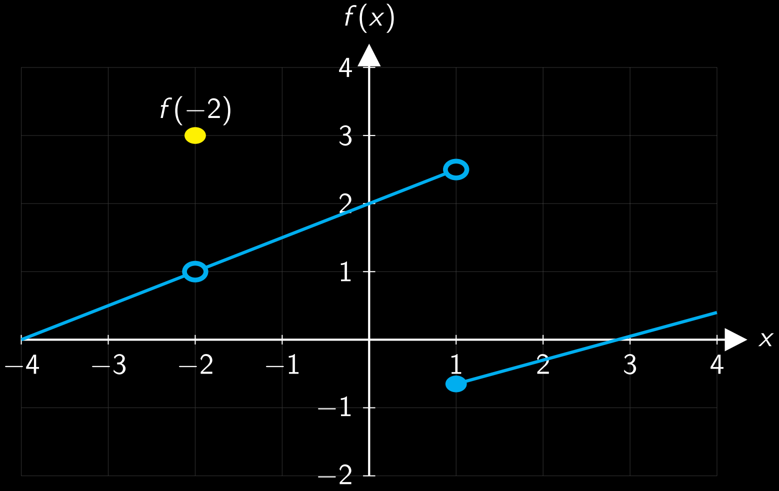
Continuous Functions

A function is continuous on an interval if it is continuous at every point in that interval.

When inspecting a graph, look for:

- holes in the graph (gaps in the domain)
- jumps in graph
- where you would need to lift your pen to draw the graph

Where Is This Function Discontinuous?



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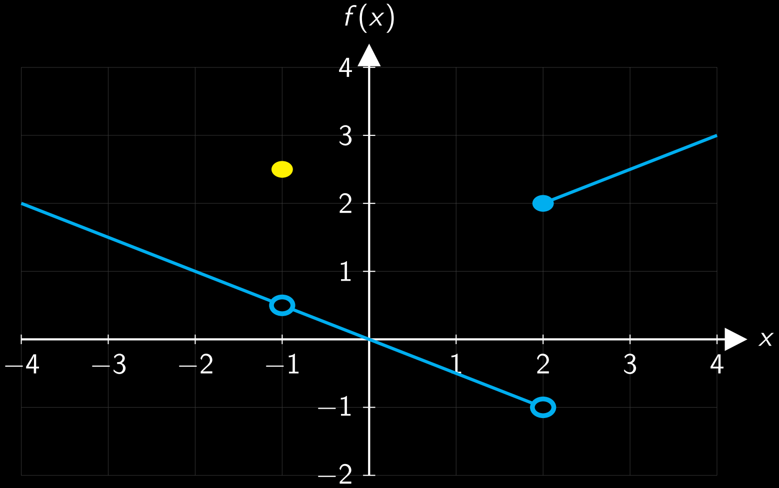
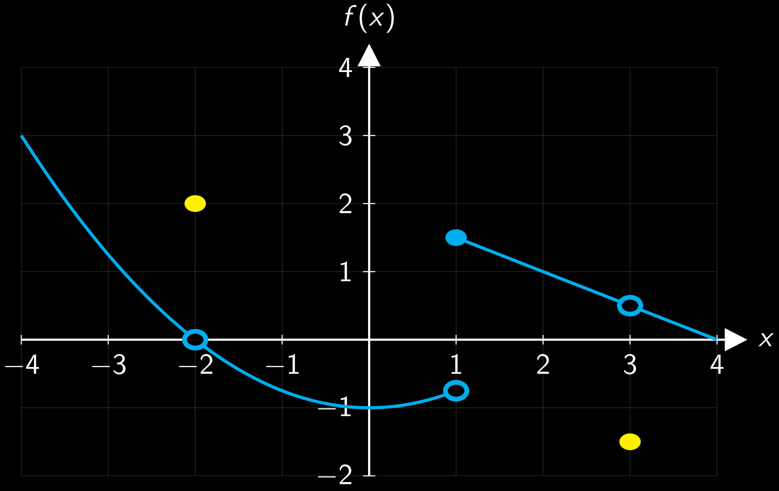


Where Is This Function Discontinuous?



Checking Continuity

$$f(x) = \begin{cases} x^2 + 2, & x \leq 2, \\ 3x, & x > 2. \end{cases}$$

$$f(2) = 2^2 + 2 = 6.$$

$$\begin{aligned} \lim_{x \to 2^-} f(x) &= \lim_{x \to 2^-} (x^2 + 2) \\ &= 2^2 + 2 \\ &= 6. \end{aligned}$$

$$\begin{aligned} \lim_{x \to 2^+} f(x) &= \lim_{x \to 2^+} (3x) \\ &= 3(2) \\ &= 6. \end{aligned}$$

Therefore, $f(x)$ is continuous at $x = 2$.

Squeeze Theorem

Suppose that, near $x = a$,

$$g(x) \leq f(x) \leq h(x).$$

If

$$\lim_{x \to a} g(x) = L \qquad \text{and} \qquad \lim_{x \to a} h(x) = L,$$

then

$$\lim_{x \to a} f(x) = L.$$

Consider This Limit

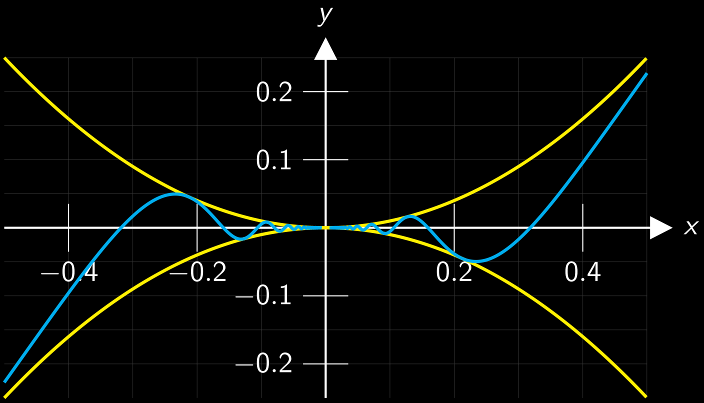
What happens to

$$f(x) = x^2 \sin\left(\frac{1}{x}\right)$$

as x approaches 0?

- $\sin(1/x)$ oscillates rapidly near $x = 0$.
- Can we trap (squeeze) the function between two other functions?

Squeezing an Oscillating Function



$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2 \quad \Longrightarrow \quad \lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

Squeeze Theorem Example

Evaluate

$$\lim_{x \to 0} x^2 \cos\left(\frac{5}{x}\right).$$

$$-x^2 \leq x^2 \cos\left(\frac{5}{x}\right) \leq x^2.$$

$$\lim_{x \to 0}(-x^2) = 0 \qquad \text{and} \qquad \lim_{x \to 0} x^2 = 0.$$

$$\therefore \quad \lim_{x \to 0} x^2 \cos\left(\frac{5}{x}\right) = 0.$$

Sigma Notation

Sigma notation gives a compact way to describe a sum:

$$\sum_{k=a}^{n} f(k).$$

- k is the index of summation.
- a is the starting value.
- n is the final value.
- $f(k)$ is the function evaluated for each value of k.

Evaluating a Sum

Evaluate

$$\sum_{k=1}^{3}(2k+1).$$

$$\begin{aligned}\sum_{k=1}^{3}(2k+1) &= (2\times 1+1)+(2\times 2+1)+(2\times 3+1)\\ &= 3+5+7\\ &= 15.\end{aligned}$$

Evaluating a Sum

Evaluate

$$\sum_{j=0}^{3} (2^j).$$

$$\sum_{j=0}^{3} (2^j) = (2^0) + (2^1) + (2^2) + (2^3)$$

$$= 1 + 2 + 4 + 8$$

$$= 15.$$

For next week

Revise derivatives of basic functions (Table of Derivatives)

Ensure that you can calculate the derivative $\left(\frac{dy}{dx}\right)$ of the following:

- $y = c \quad$ (c is a constant)
- $y = ax^n \quad$ (a and n are constants)
- $y = \sin(x)$, $y = \cos(x)$
- $y = e^x$
- $y = \ln(x)$